Helicopter vs Car Travel Time Analysis

A Comprehensive Comparison of Traffic Scenarios
New York & Los Angeles

REVO

December 2025

Agenda

The Problem: Urban Traffic Congestion

The Challenge

- Business travelers need reliable airport access
- Traffic is unpredictable and often severe
- Missing flights = significant business impact
- Stress and time loss affect productivity

Worst Case Reality

During severe traffic:

NY: 5.20x travel time

LA: 4.15x travel time

12-17 km/h avg speed

(barely faster than walking!)

Study Overview

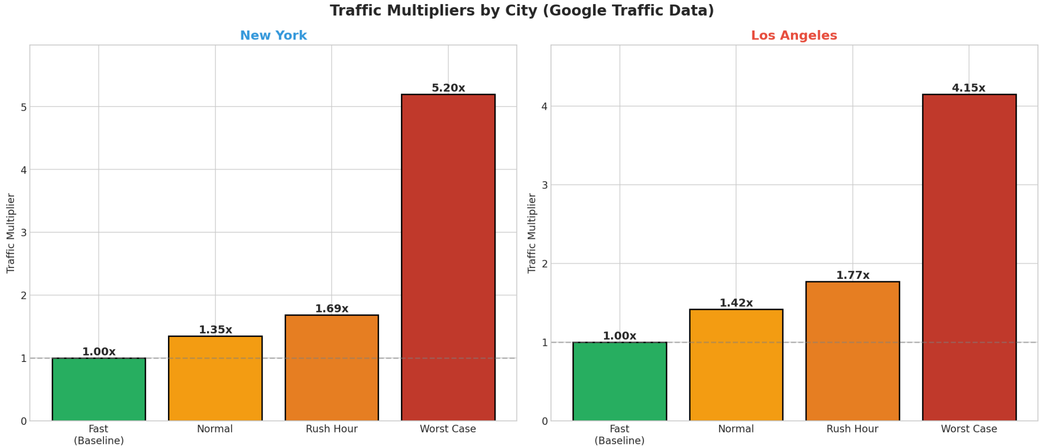
Scope

- **New York**: 70 wealthy ZIP codes → JFK
- **Los Angeles**: 44 wealthy ZIP codes → LAX
- **Data**: Google Traffic (Pessimistic)
- **Routing**: OSRM + FAA Helipad Data

Key Numbers

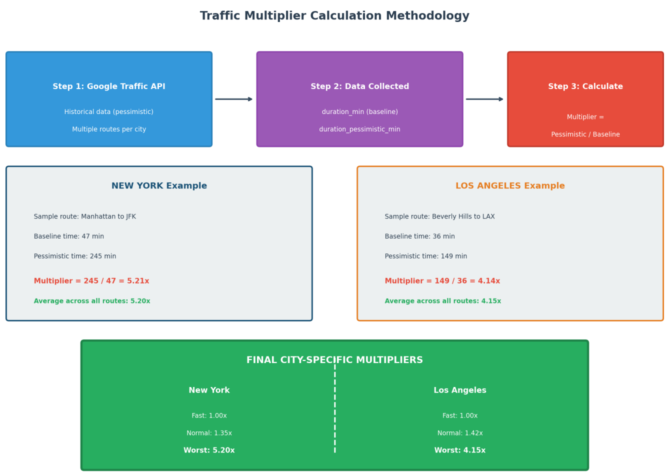
114 ZIP Codes
65 Heliports
4 Traffic Scenarios
27 Manhattan ZIPs

City-Specific Traffic Multipliers

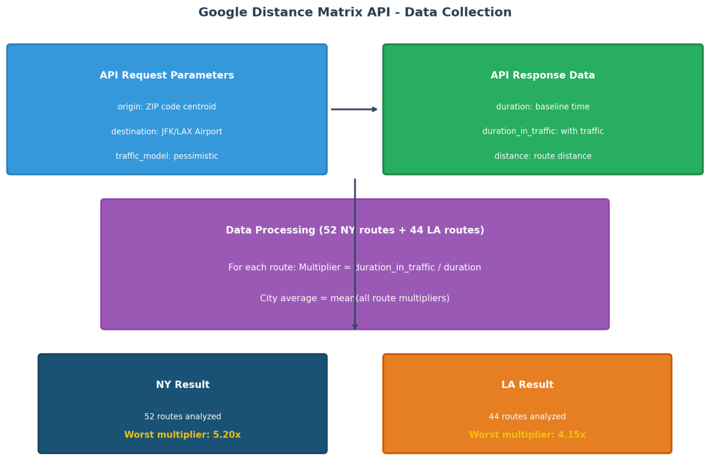


NY: Higher worst case (5.20x) | LA: Higher normal traffic (1.42x)

How We Calculate Traffic Multipliers



Google Traffic API: Data Flow



Multiplier Calculation: Numerical Examples

Route: Manhattan → JFK

Baseline:	47 min
Pessimistic:	245 min
Multiplier:	$\frac{245}{47} = \mathbf{5.21x}$

Los Angeles Example

Route: Beverly Hills → LAX

Baseline:	36 min
Pessimistic:	149 min
Multiplier:	$\frac{149}{36} = \mathbf{4.14x}$

NY City Average

52 routes analyzed

Final: 5.20x

LA City Average

44 routes analyzed

Final: 4.15x

Data source: Google Distance Matrix API with traffic_model=pessimistic

Why Pessimistic Data?

Google Traffic Models

- **best_guess**: Average conditions
- **optimistic**: Lighter traffic
- **pessimistic**: Worst historical

Our Choice: Pessimistic

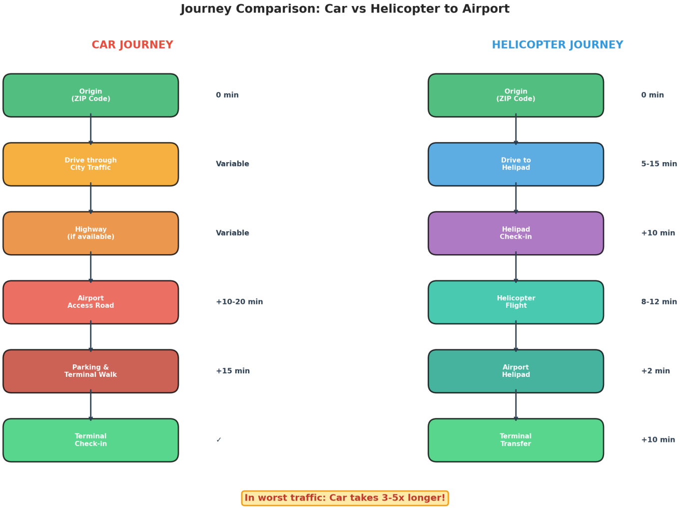
- Represents real worst-case scenarios
- Based on actual historical data
- Critical for risk assessment

What Causes Worst Traffic?

- Major accidents
- Severe weather
- Special events (sports, concerts)
- Holiday periods
- Road construction

Result: NY 5.20x | LA 4.15x travel time in worst conditions

Journey Comparison: Car vs Helicopter



Helicopter Journey Components

Step 1: Drive to Helipad

- Affected by traffic
- Manhattan helipad priority
- Average: 5-15 minutes

Step 2: Check-in

- Fixed 10 minutes
- Security screening
- Boarding preparation

Step 3: Flight

- 200 km/h cruise speed
- Direct route
- 8-12 minutes typical

Step 4: Terminal Transfer

- Fixed 10 minutes
- Helipad to terminal
- Internal transport

Manhattan Helipad Network

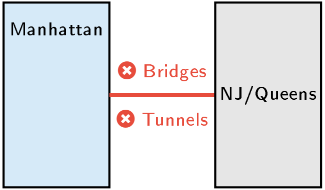
Approved Heliports

- **JRB**: Downtown/Wall St
- **JRA**: West 30th Street
- **6N5**: East 34th Street
- **3NY2**: Astoria (Queens)

Exclusions

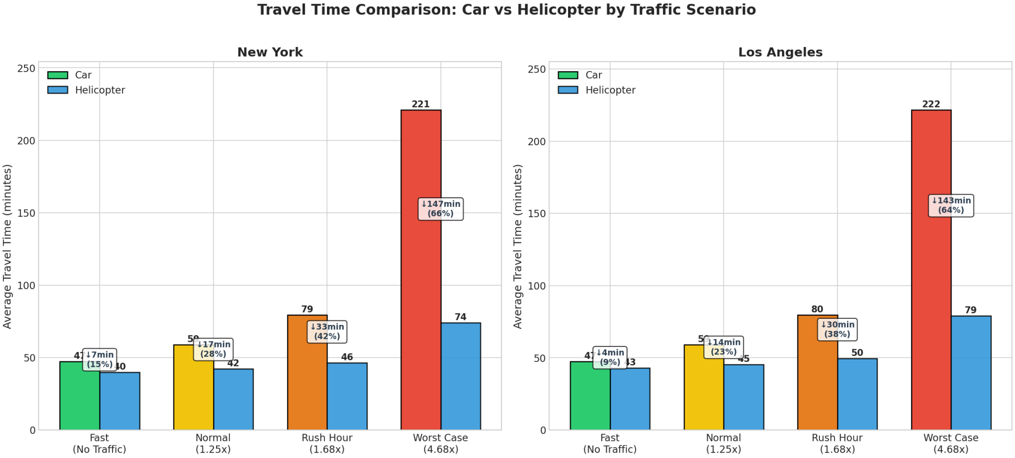
- Police Heliports
- One Police Plaza
- Newport Beach Police (LA)

Why Manhattan-Only?



Avoid congested crossings!

Scenario Comparison Overview

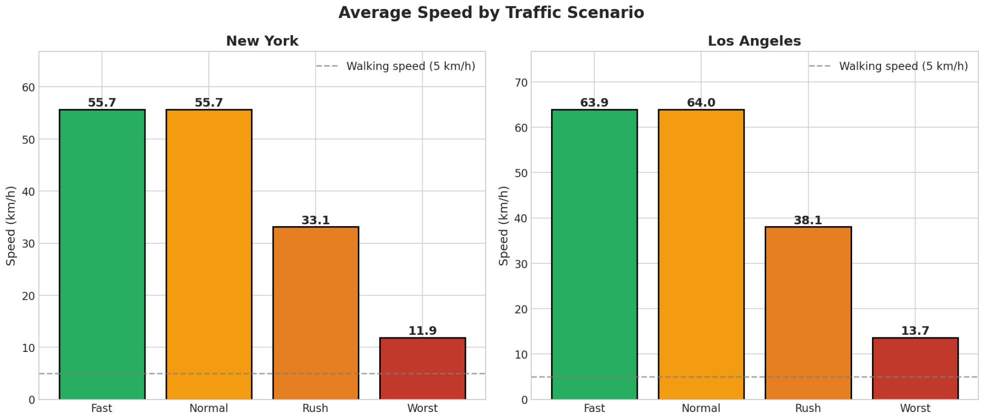


Key Statistics

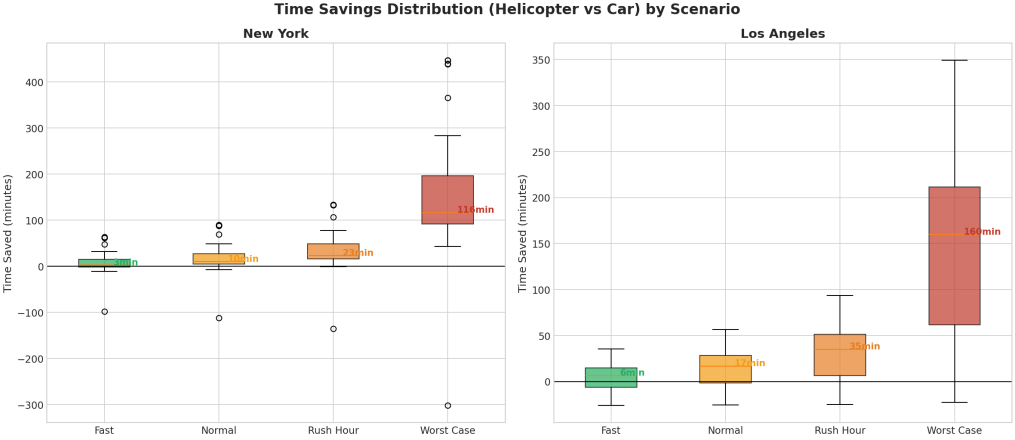
Metric	New York Fast	New York Norm	New York Rush	New York Worst	Los Angeles Fast	Los Angeles Norm	Los Angeles Rush	Los Angeles Worst
Car (min)	47	59	79	**221**	47	59	80	**222**
Heli (min)	44	46	49	**74**	46	48	52	**79**
Savings	3	13	30	**147**	1	11	28	**143**

Worst Case: Save 2+ Hours!

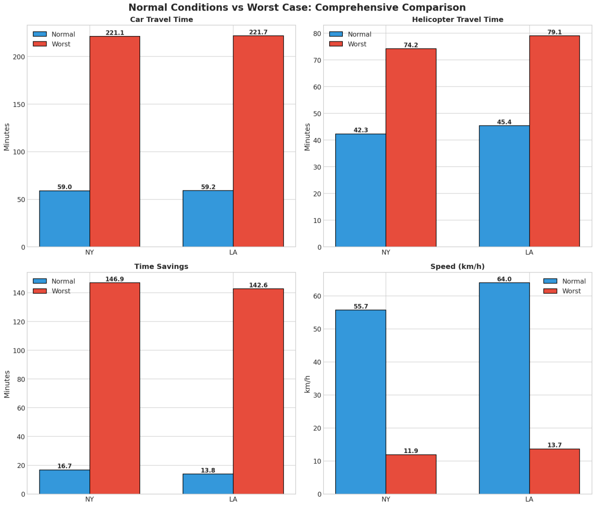
Speed Degradation by Scenario



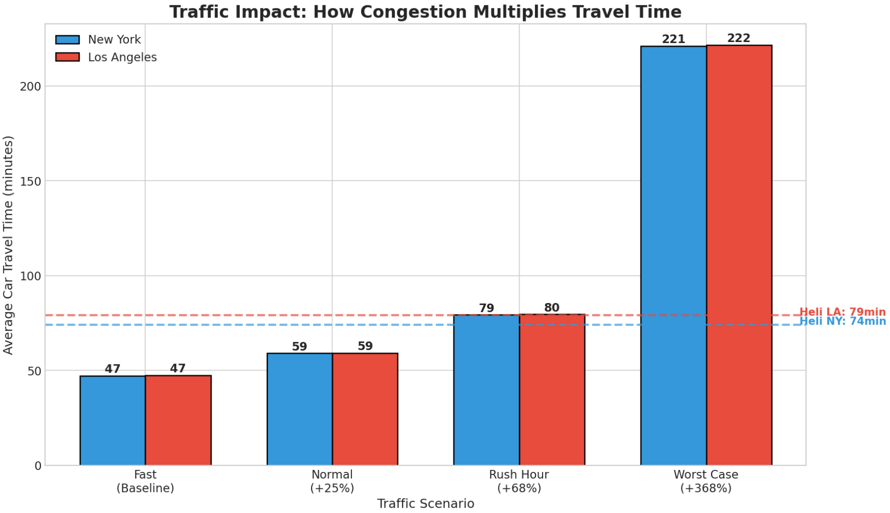
Time Savings Distribution



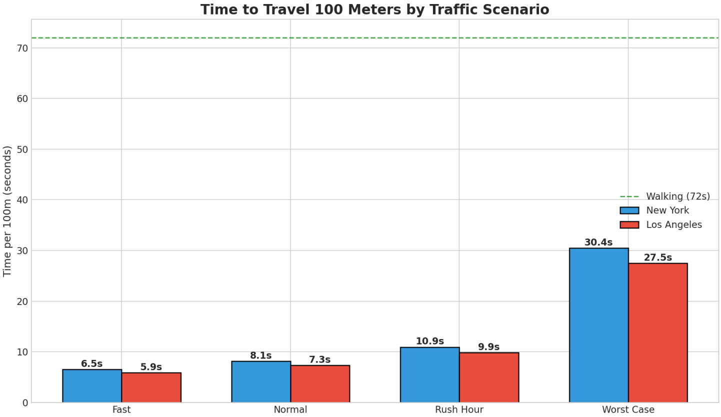
Normal vs Worst Case: The Dramatic Difference



How Traffic Multiplies Travel Time



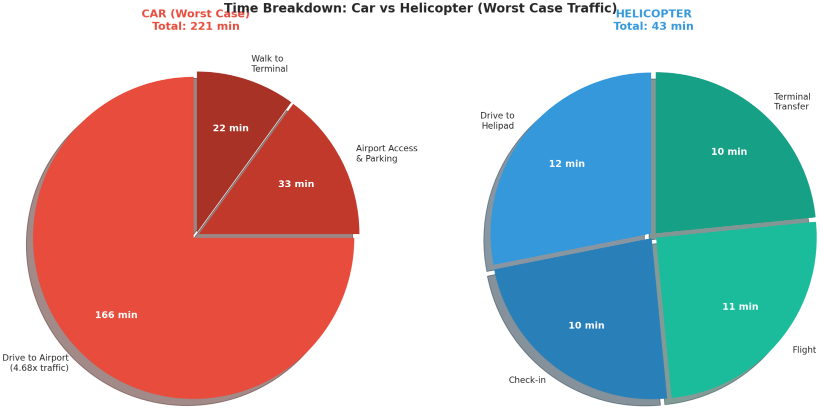
Time per 100 Meters



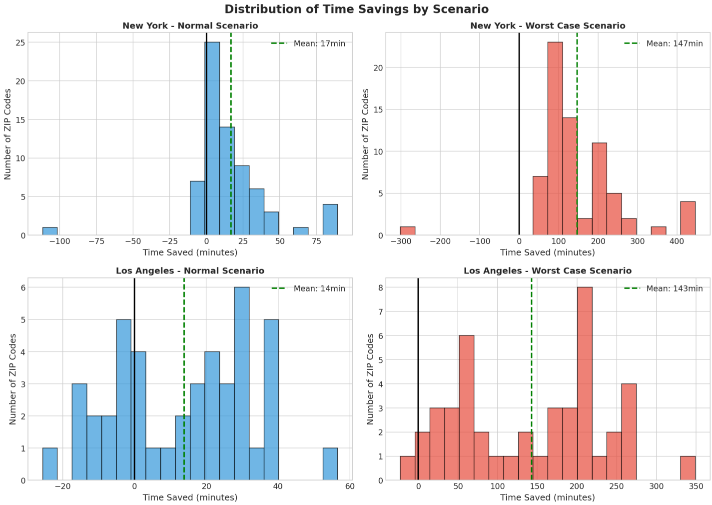
Worst Case: 30 sec / 100m

Walking: 72 sec / 100m

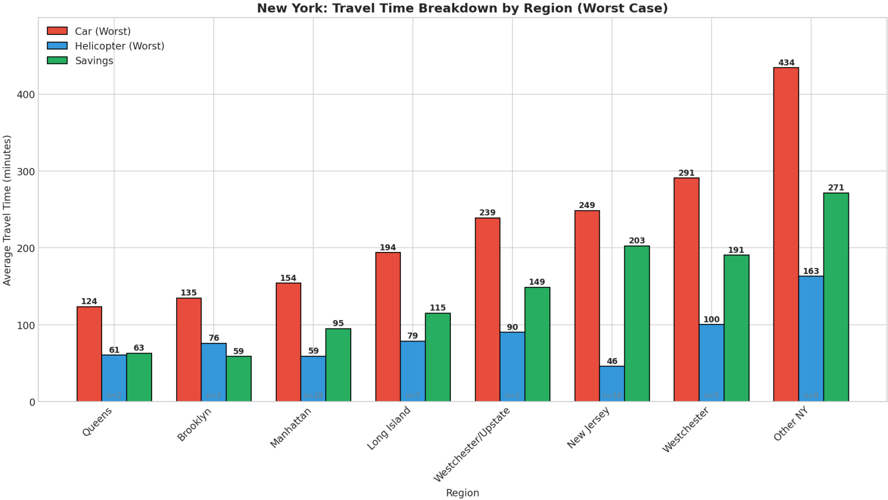
Time Breakdown: Worst Case



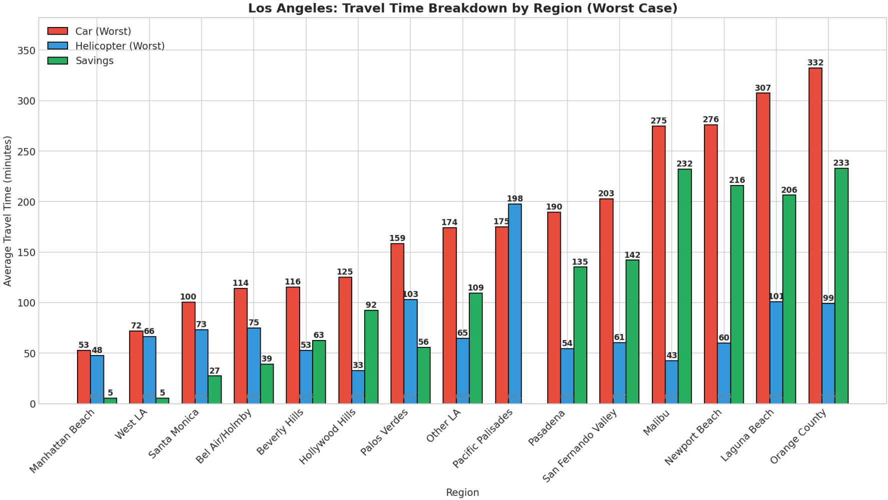
Savings Distribution Histogram



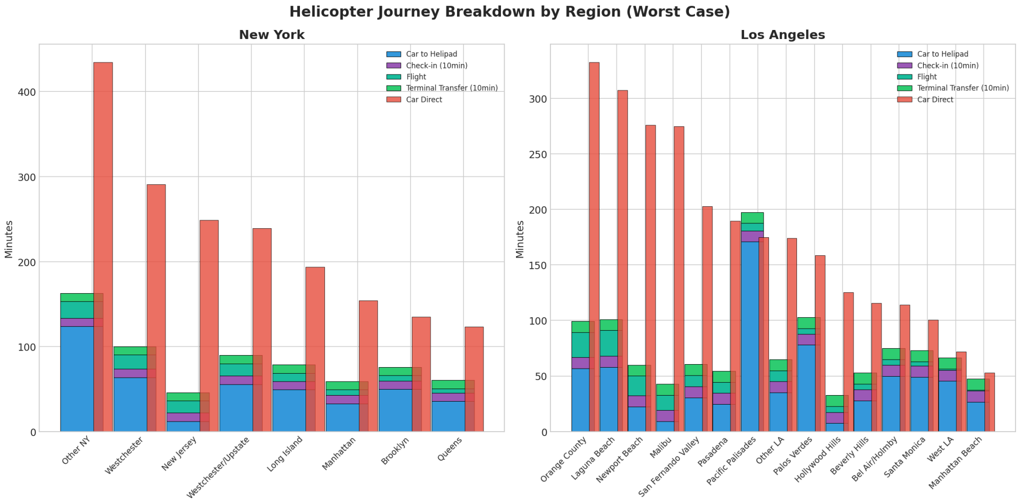
New York: Regional Breakdown



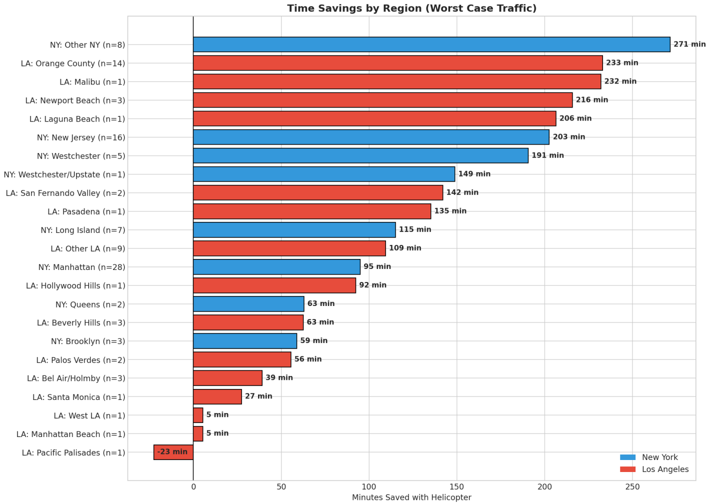
Los Angeles: Regional Breakdown



Journey Components by Region



Regional Savings Summary



Interactive Maps with OSRM Routes

New York Map

- 70 ZIP codes analyzed
- 27 in Manhattan
- Real OSRM driving routes
- Routes appear on hover

`map_ny_osrm.html`

Los Angeles Map

- 44 ZIP codes analyzed
- Multiple regions covered
- Real OSRM driving routes
- Routes appear on hover

`map_la_osrm.html`

Hover over ZIP codes to see: Blue route to helipad | Red route to airport

Map Features: Route Visualization

Route Colors

- **Blue dashed line:** Route to nearest helipad
- **Red dashed line:** Route to main airport (JFK/LAX)

ZIP Code Colors (Savings)

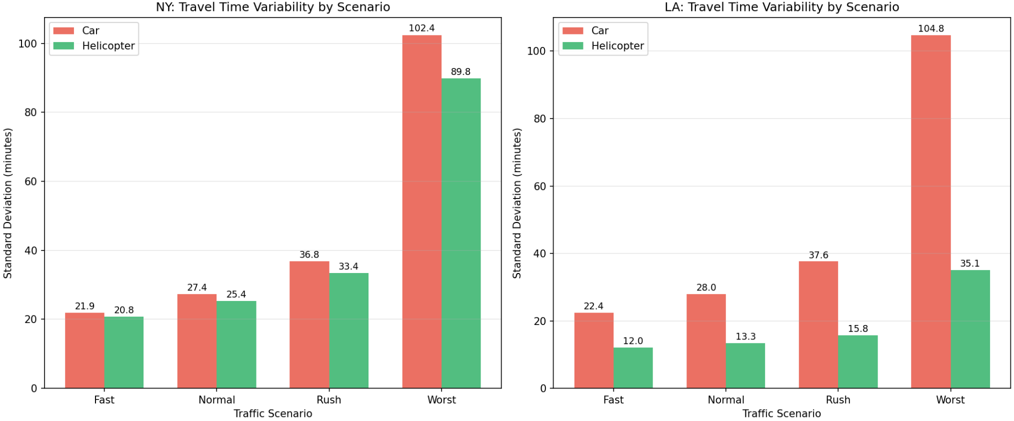
- **Dark green:** > 100 min savings
- **Green:** 50-100 min savings
- **Orange:** 0-50 min savings
- **Red:** No savings

Info Panel Shows

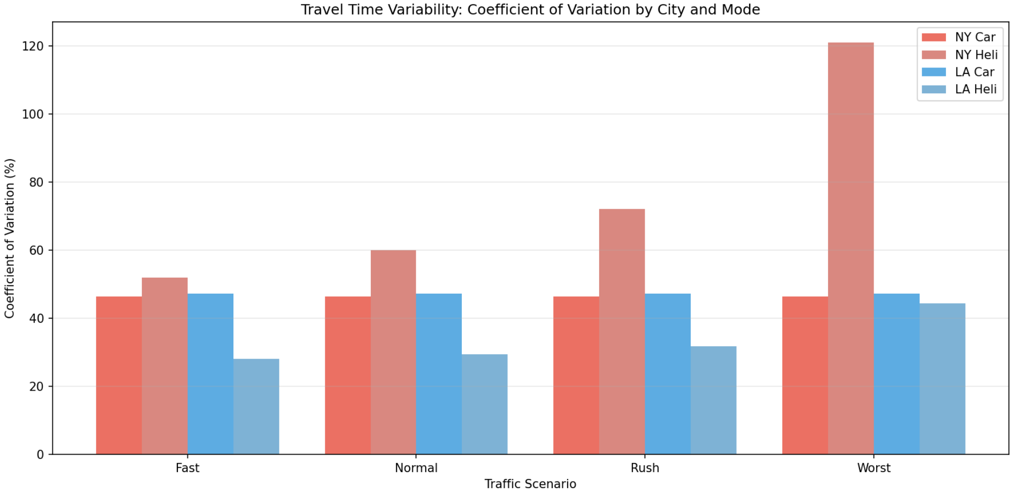
- Nearest helipad name
- Distance in km
- Flight time
- Car times: Fast/Rush/Worst
- Heli times: Fast/Rush/Worst
- **Total savings**

Access: `victorcandido1.github.io/cityexpansions_corporate/v1/`

Travel Time Variability by Scenario



Coefficient of Variation: Predictability



Regional Variance: Key Findings

Most Predictable (Low CV)

- **Pasadena**: CV = 4.9%
- **Manhattan**: CV = 7.9%
- Beverly Hills: CV = 21.7%

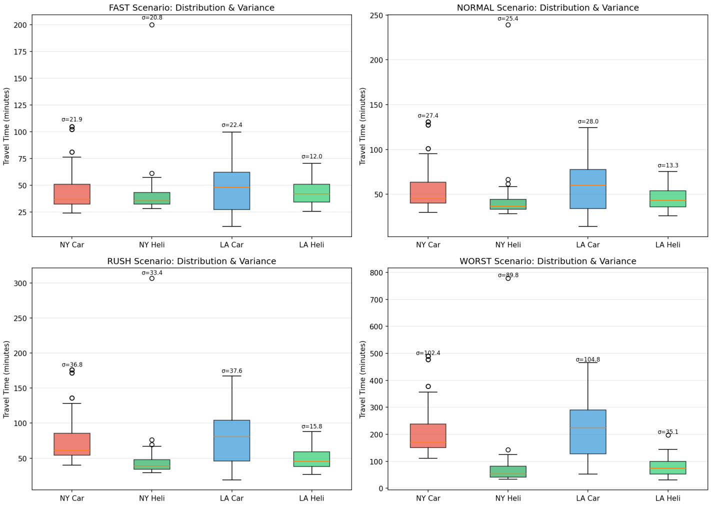
Least Predictable (High CV)

- **Long Island**: CV = 47.8%
- **Orange County**: CV = 35.4%
- Malibu: CV = 32.2%

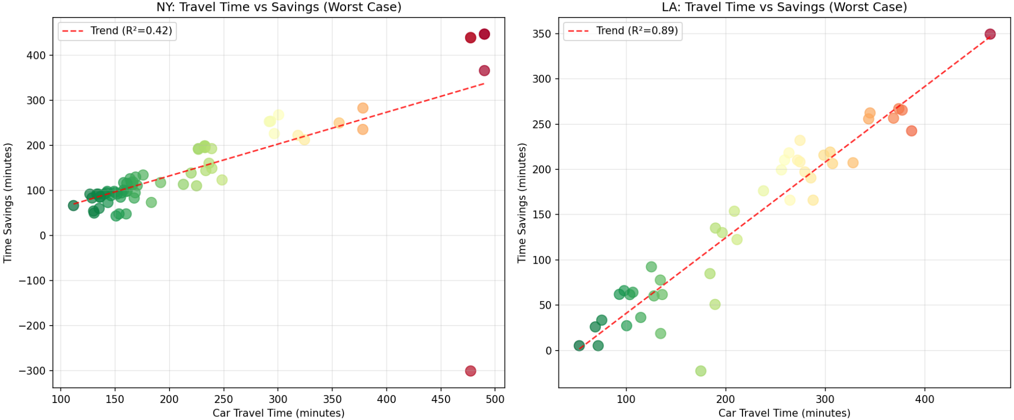
What This Means

Low CV = More predictable travel times = Lower risk of delays
High CV = Higher uncertainty = Greater need for helicopter reliability

Distribution Analysis (Boxplots)



Travel Time vs Savings Correlation



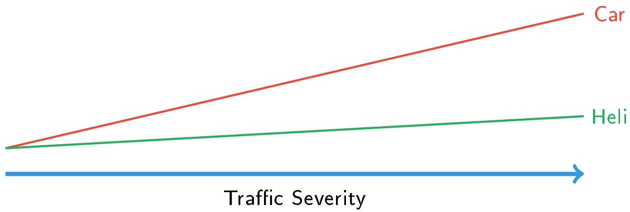
Why Helicopter Wins in Worst Case

Car Travel

- **100%** affected by traffic
- NY: Time increases **5.20x**
- LA: Time increases **4.15x**
- Unpredictable delays

Helicopter Travel

- Only **20%** affected by traffic
- Time increases only **1.5-1.8x**
- Predictable schedule
- Comfort and productivity



Business Value Proposition

Time Savings

147 min
(NY worst case)

Reliability

99%
advantage rate

Never Miss

Flights
predictable arrival

ROI Consideration

2+ hours saved = productive meeting time, reduced stress, guaranteed connections

Summary of Findings

	Los Angeles
70 ZIP codes analyzed	44 ZIP codes analyzed
27 Manhattan locations	Distributed coverage
147 min worst case savings	**143 min** worst case savings
66% time reduction	**64%** time reduction
99% show helicopter advantage	98% show helicopter advantage

Key Takeaway

The worse the traffic, the greater the helicopter advantage.
In extreme conditions, helicopter is the **only viable option** for time-sensitive travel.

Complete Summary Statistics

Comprehensive Scenario Comparison Summary

Metric	NY Fast	NY Normal	NY Rush	NY Worst	LA Fast	LA Normal	LA Rush	LA Worst
Car Time (min)	47	59	79	221	47	59	80	222
Heli Time (min)	40	42	46	74	43	45	50	79
Savings (min)	7	17	33	147	4	14	30	143
Savings (%)	15%	28%	42%	66%	9%	23%	38%	64%
Speed (km/h)	55.7	55.7	33.1	11.9	63.9	64.0	38.1	13.7
Time/100m (sec)	6.5	8.1	10.9	30.4	5.9	7.3	9.9	27.5
% Heli Advantage	56%	87%	97%	99%	59%	68%	82%	98%

Thank You

Questions?

REVO Helicopter Services
Strategic Analysis Division

December 2025